GSE178341 reference: 71 unmerged subtype candidates (at least 50 cells)
Frozen cell definitions and source association directions

scRNA primary branch
Historical source cutoff >0.65: 24 candidates
Historical configuration comparison: fixed18
18 proportions + weighted malignant MLH1

Grouped classifier OOF + retrospective scoring
Training-only Z-score + shrinkage LDA
Frozen18 model transferred to spatial proxies

Selection stability: 2,000 development-only resamples
Source screen and measurements fixed
Panel adoption includes post hoc configuration comparison

Bulk candidate branch
Source oriented AUROC >0.70: 17 candidates
Both development oriented AUROCs >0.50
12 direction-concordant subtypes

GSE39582 + TCGA-COADREAD
Archived five-fold OOF / full-fit MLP
Training-only Z-score, fixed 4-unit MLP

MMR4: fit 7 cohorts, score 1 held-out cohort
Fixed fusion: 0.5 subtype probability + 0.5 MMR4
Two development + six retrospective cohorts